

# Tyler Lammey

[tylerlammey@gmail.com](mailto:tylerlammey@gmail.com) | [github.com/tylerlammey](https://github.com/tylerlammey) | [linkedin.com/in/tyler-lammey](https://linkedin.com/in/tyler-lammey) | [tylerlammey.com](https://tylerlammey.com)

**Security Clearance:** Active DoD Secret Clearance

## EDUCATION

### Rensselaer Polytechnic Institute

Troy, NY

*Bachelor of Science in Computer and Systems Engineering (Computer Engineering)*

*Sept. 2023 – May 2027*

- GPA: 3.82 | Dean's Honor List (All Semesters)
- Relevant Coursework: Data Structures, Algorithms, Foundations of CS, Computer Architecture, Computer Networks, Network Security, Embedded Control, Computational Vision, Signals & Systems, Quantum Devices
- Clubs: RPai (AI Club), RPISEC (Cybersecurity Club), Quantum Computing Club

## WORK EXPERIENCE

### Raytheon (RTX)

Tewksbury, MA

*Radar Systems Engineering Intern (LTAMDS), Modeling and Simulation*

*May 2026 – Aug. 2026*

- Debugged MATLAB codebase by identifying incorrect variables within `.mat` data structures through exploratory plotting and validation, correcting defects and documenting fixes for future analysis.
- Developed a VBA automation tool integrated with Excel and Word to generate PDF deliverables from 100+ records, reducing over 6 hours of manual processing per run while eliminating copy errors.
- Validated software updates through structured testing and created technical documentation detailing data structure corrections and workflow improvements for engineering teams.

### Scientific Research Corporation

Huntsville, AL

*Radar Engineer Intern, Waveform Generation*

*May 2025 – Aug. 2025*

- Developed Python automation tools to communicate with SCPI-controlled signal analyzers, streamlining RF measurement procedures and simplifying waveform testing procedures for engineers.
- Created a Python-based waveform conversion interface that eliminated manual waveform regeneration when transitioning between software-defined radio (SDR) file formats, improving flexibility of testing workflows.
- Collaborated with engineers to validate waveform generation and SDR performance using laboratory signal analysis equipment, documenting tools and procedures for future users.

### Advanced Technologies and Services, Inc.

Wyckoff, NJ

*Software Engineer Intern*

*May 2024 – Aug. 2024*

- Automated IoT device production using Arduino firmware and Linux Bash scripts, reducing device preparation time by over 50% while minimizing manual setup errors and improving yields.
- Developed Python-based TCP/UDP network testing tools using Python and Linux sockets and deployed testing frameworks on Google Cloud infrastructure to evaluate distributed IoT device performance.
- Contributed to a web-based ISP network management platform using Angular, FastAPI, and cloud services, supporting backend API development and device communication workflows.

## PROJECTS

### TylerGPT (AI Portfolio Agent) | *Python, FastAPI, React, OpenAI API*

- Developed and deployed a full-stack AI portfolio assistant using React and FastAPI with OpenAI tool calling, enabling automated contact capture and unanswered-question tracking.
- Engineered an asynchronous LLM streaming backend with recursive tool execution, conversational context management, REST APIs, Docker deployment, and automated GitHub CI/CD across Vercel and Railway.

### Automated IoT Plant Care System | *C++, Python, FastAPI, React, Arduino*

- Designed the full-stack architecture for a six-person team project, independently developing firmware, backend services, frontend components, and circuit integration.
- Built Arduino firmware integrating temperature, soil moisture, and water-level sensors with relay-controlled actuators, and built a React dashboard with FastAPI REST APIs for remote monitoring and control.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, JavaScript, TypeScript, HTML/CSS

**Frameworks & Libraries:** React, FastAPI, Angular, PyTorch, Hugging Face, OpenCV, NumPy, Pandas, Matplotlib

**AI/ML:** Deep Learning, Object Detection & Segmentation (SAM), LLM Tool Calling, Prompt Engineering

**Tools & Embedded:** Git, Linux, Docker, REST APIs, MATLAB, Arduino, Ethernet, SPI, PWM, Jupyter Notebooks